



LUGGAGE TRACKER AND MONITORING SYSTEM



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1. OVERVIEW :

The Smart Luggage Tracking System is designed around the ESP32 microcontroller, which acts as the central processing and communication unit. Multiple sensors and modules are interfaced with the ESP32 to provide location tracking, environmental monitoring, motion detection, and user alert functionality.

The system integrates GPS tracking, motion sensing, environmental sensing, local feedback, and cloud-based notifications to ensure luggage safety and real-time monitoring.

2. Problem Statement

Luggage loss and theft during travel are common issues, and existing tracking solutions are either costly or lack real-time monitoring. There is a need for an affordable, smart system that can track luggage location, detect unauthorized movement, and provide remote monitoring through a web-based platform.

3. Objectives

- To design a smart luggage tracking system using ESP32
- To track luggage location in real time using GPS
- To detect unauthorized motion and bag opening
- To monitor temperature and humidity inside the luggage
- To display real-time data on the Blynk Web Dashboard

4. Hardware Components and Their Roles

4.1 ESP32 Microcontroller

ESP32 is the core controller of the system. It collects data from all connected sensors, processes the information, and sends relevant alerts to the user via the Blynk IoT application using Wi-Fi connectivity.

4.2 GPS Module (NEO-6M)

The GPS module is responsible for fetching the real-time geographical location of the luggage in terms of latitude and longitude. This location data is continuously monitored and shared with the user through the mobile application.

4.3 PIR Motion Sensor

The PIR (Passive Infrared) sensor detects human motion around the luggage. If unexpected movement is detected, it triggers the alert system.

4.4 DHT11 Sensor

The DHT11 sensor measures temperature and humidity inside the luggage. This helps in monitoring environmental conditions, especially for sensitive items.

4.5 LDR Sensor

The LDR (Light Dependent Resistor) detects light intensity inside the bag. It is mainly used to determine whether the luggage is opened or closed.

4.6 OLED Display

The OLED display provides local feedback by showing system status, sensor readings, and alerts directly on the luggage device.

4.7 Buzzer Alarm

The buzzer acts as an immediate theft alert system. When suspicious activity such as motion or bag opening is detected, the buzzer is activated to alert nearby people.

5. Software and IoT Integration

5.1 Blynk Web Dashboard

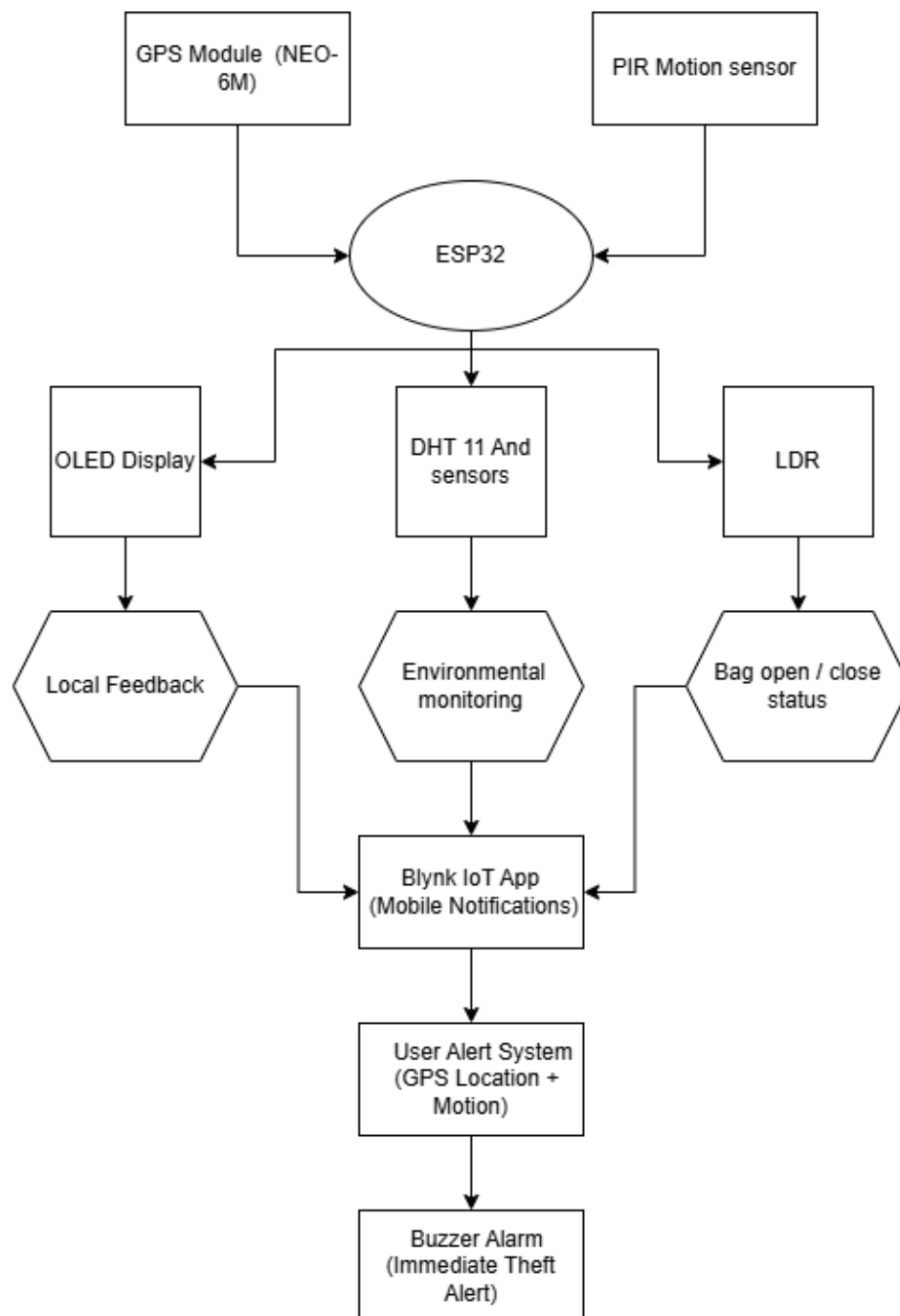
The Blynk Web platform is used for cloud-based monitoring and alert management of the Smart Luggage Tracking System. The ESP32 sends sensor data and GPS coordinates to the Blynk Cloud using Wi-Fi. This data is then visualized on the Blynk Web Dashboard, which can be accessed through any web browser.

The dashboard provides real-time updates of luggage location, motion status, bag open/close condition, and environmental parameters such as temperature and humidity.

Functions of Blynk Web Dashboard:

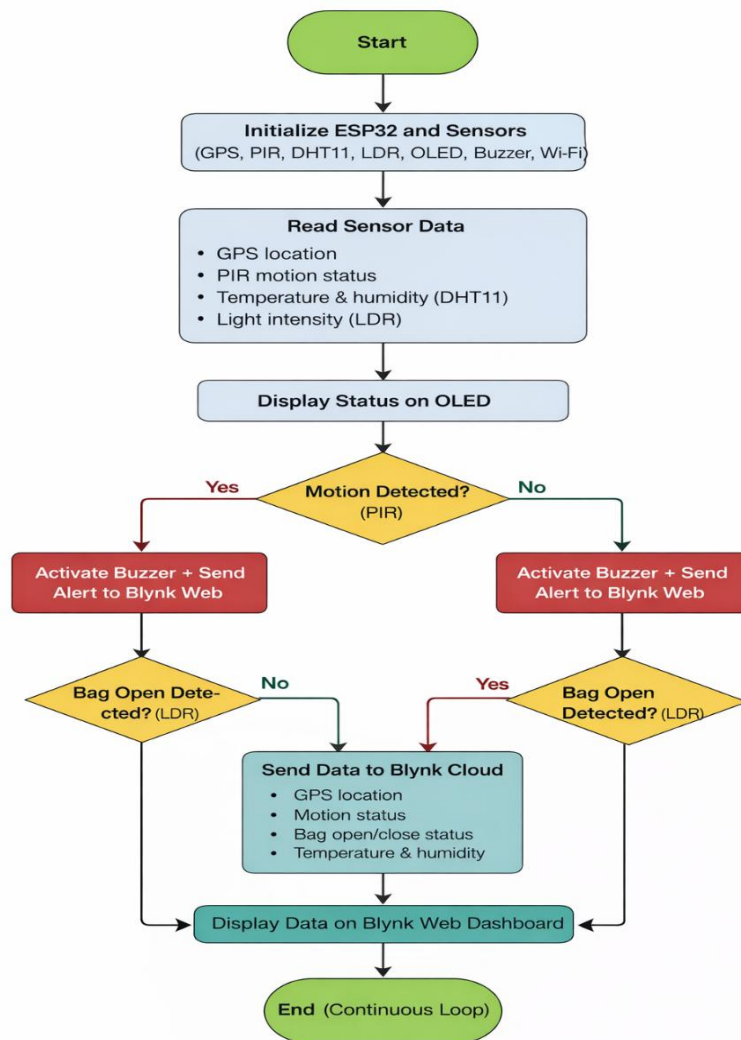
- Live GPS location monitoring through latitude and longitude
- Motion detection alerts displayed on the web interface
- Bag open/close status monitoring using LDR data
- Real-time temperature and humidity visualization
- Centralized cloud-based data access

6. Block diagram :



7. System Data Flow And Flow Chart

- The GPS module provides real-time location data to the ESP32.
- PIR sensor, DHT11, and LDR send motion, environmental, and bag open/close data to the ESP32.
- The ESP32 processes all sensor inputs and displays status on the OLED.
- If motion or tampering is detected, the buzzer is activated immediately.
- Processed data is sent to Blynk Cloud via Wi-Fi.
- The Blynk Web Dashboard displays GPS location, alerts, and sensor data for user monitoring.



8. PIN MAP:

#	Component	Signal / Pins	ESP32 GPIO	Power
1	DHT11 (Temp & Humidity)	DATA	GPIO 4	3.3V
		VCC	3.3V	
		GND	GND	
2	PIR Motion Sensor	OUT	GPIO 27	5V
		VCC	5V	
		GND	GND	
3	LDR (Light Sensor)	Analog Signal	GPIO 34 (ADC)	3.3V
		GND	GND	
4	Buzzer	+	GPIO 26	—
		—	GND	
5	GPS NEO-6M	TX	GPIO 16	3.3V / 5V
		RX	GPIO 17	
		VCC	3.3V / 5V	
		GND	GND	
6	OLED SSD1306 (I2C)	SDA	GPIO 21	3.3V
		SCL	GPIO 22	
		VCC	3.3V	
		GND	GND	

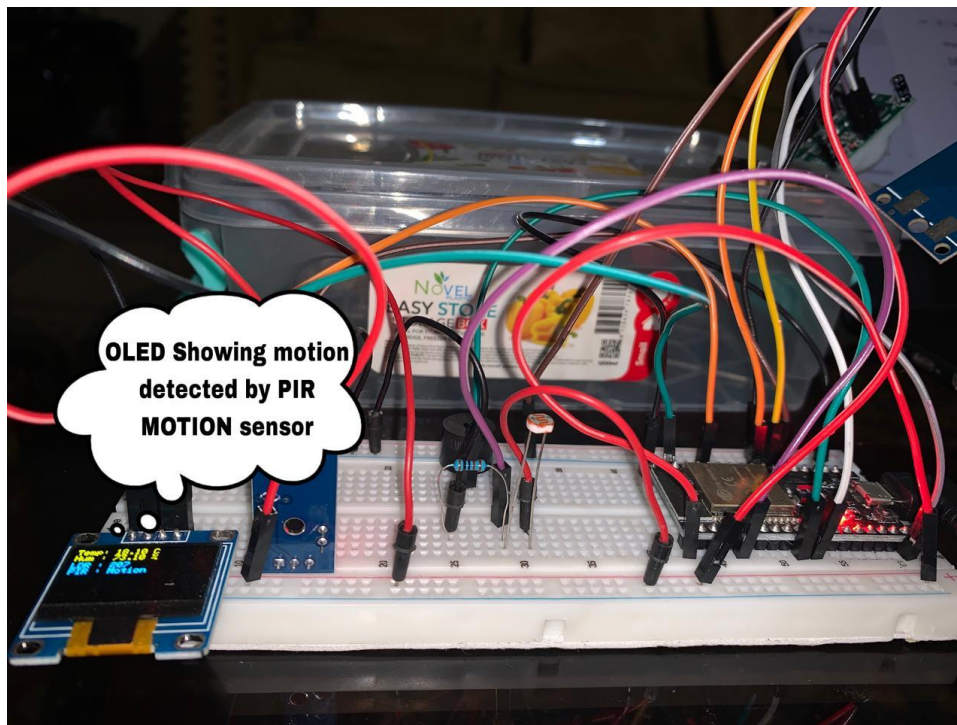
9. DATA MAP (BLYNK):

Sensor / Data	Blynk Virtual Pin	Data Type
Temperature (DHT11)	V0	Float
Humidity (DHT11)	V1	Float
LDR (Light value)	V2	Integer
PIR Motion	V3	0 / 1
GPS Latitude	V10	Double
GPS Longitude	V11	Double

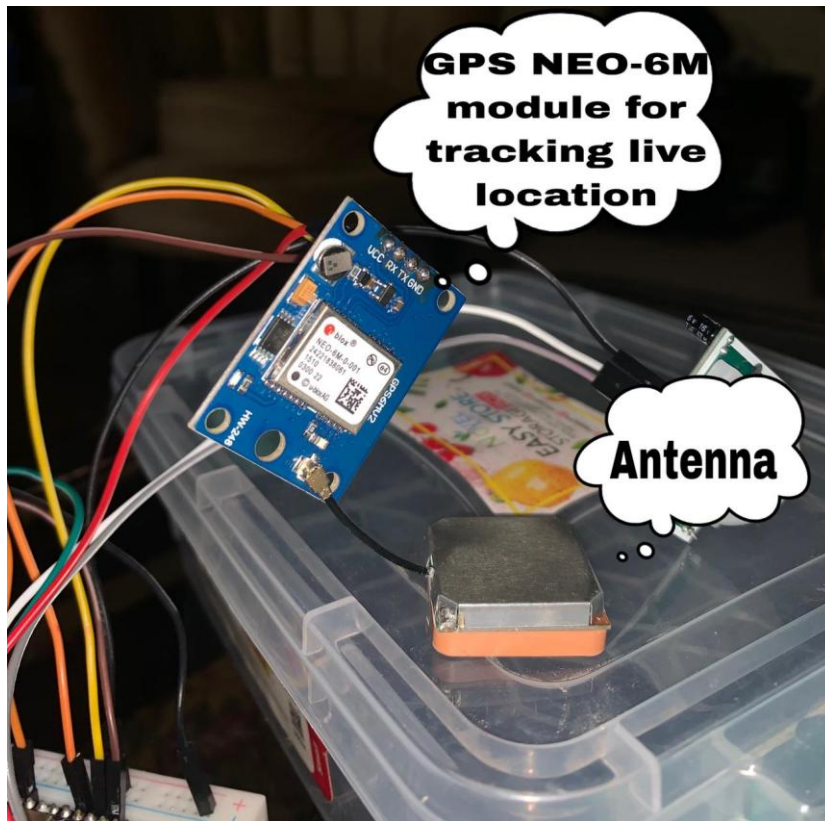
Sensor / Data	Blynk Virtual Pin	Data Type
GPS Satellites	V12	Integer
GPS Altitude (m)	V13	Double

10. CIRCUIT PICS :

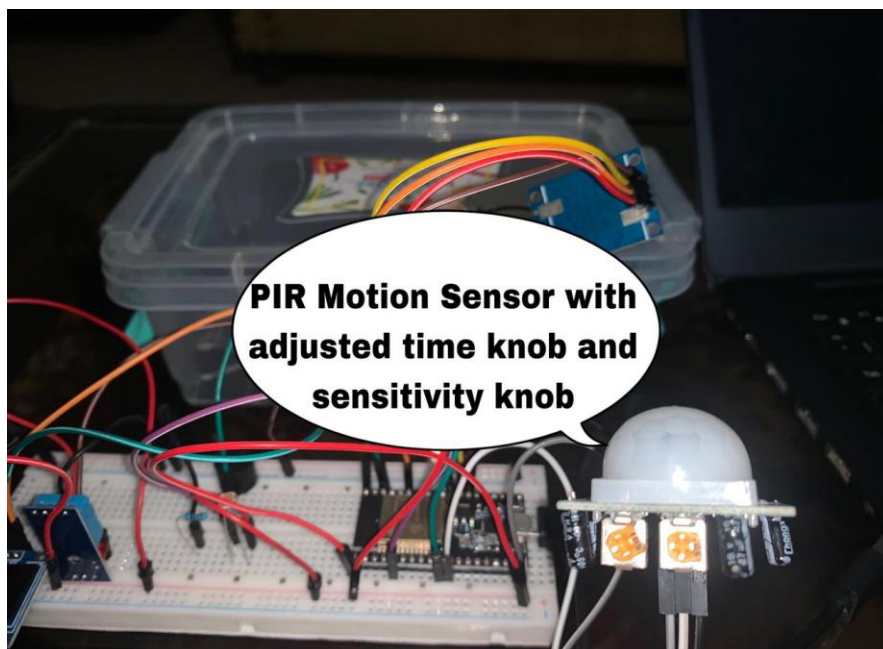
- OLED



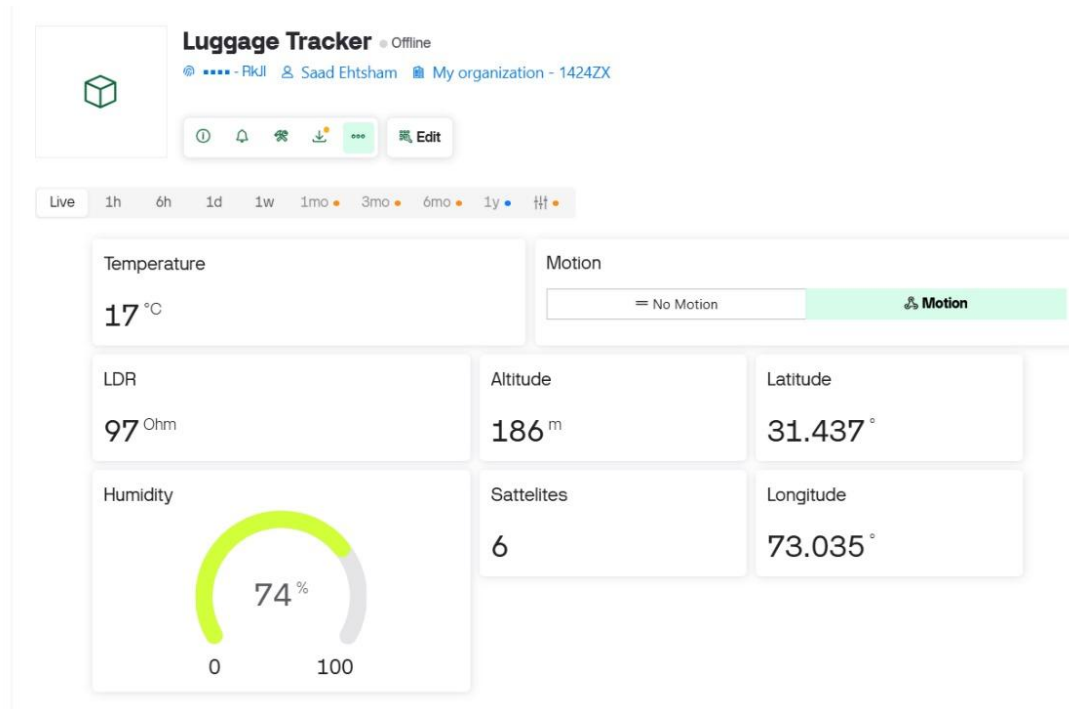
- **GPS MODULE**



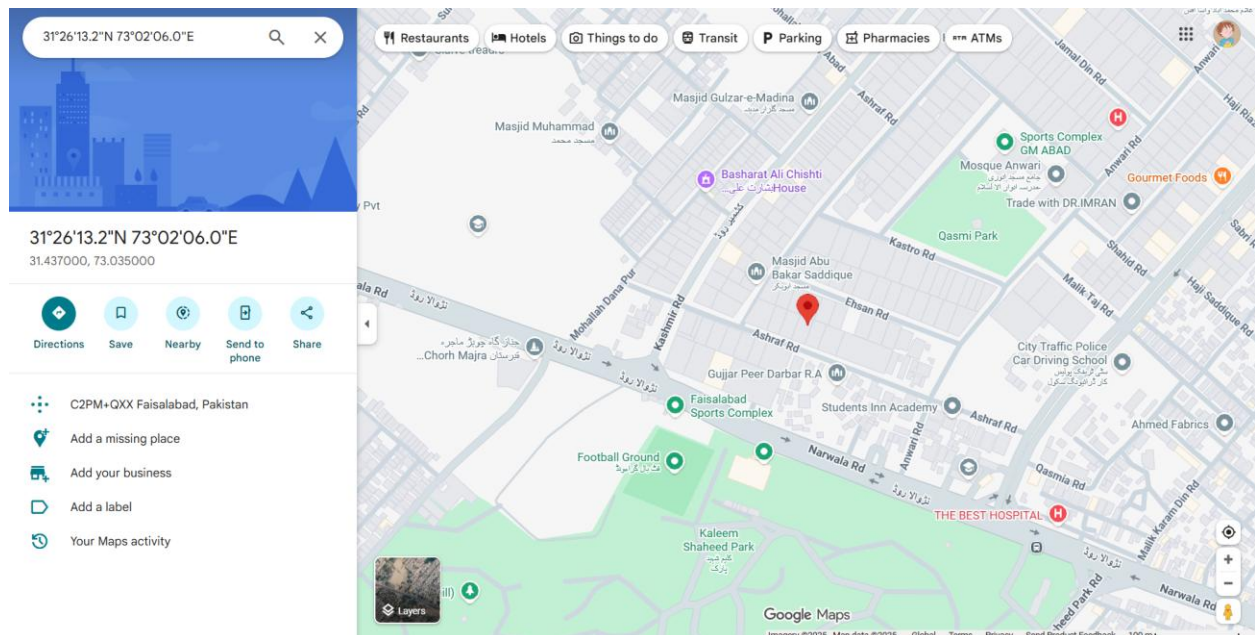
- **PIR MOTION SENSOR**



11. BLYNK DASHBOARD :



COORDINATES ON GOOGLE MAPS:



12. Results

The Smart Luggage Tracking and Monitoring System was successfully implemented and tested. The ESP32 accurately collected data from the GPS, PIR, DHT11, and LDR sensors. Real time location data was reliably transmitted to the Blynk Cloud and displayed on the Blynk Web Dashboard. Motion detection and bag opening events were correctly identified, triggering the buzzer for immediate local alerts. Environmental parameters such as temperature and humidity were monitored effectively. Overall, the system demonstrated stable performance, timely alerts, and accurate data visualization through the web interface.

13. Conclusion

The Smart Luggage Tracking and Monitoring System provides an efficient and low cost solution for luggage security using IoT technology. By integrating GPS tracking, motion detection, environmental monitoring, and web based data visualization, the system ensures real-time monitoring and theft prevention. The use of the Blynk Web Dashboard allows platform-independent access without the need for a dedicated mobile application. This project successfully demonstrates the practical application of embedded systems and IoT in solving real world travel security problems.

14. Future Scope

The system can be further enhanced with additional features to improve functionality and reliability. Future improvements may include geo fencing alerts to notify users when luggage moves beyond a defined area, GSM based communication as a backup for WiFi failure, and battery optimization for longer usage. Integration of a camera module for visual monitoring and cloud data analytics for movement history can also enhance system performance. These enhancements would make the system more robust and suitable for commercial deployment.